

MA 102 Mathematics II
Ordinary Differential Equations
Lecture 13
(April 3, 2024)

Important recurrence relations:

$$(1) \quad \frac{d}{dx} (x^\mu J_\mu(x)) = x^\mu J_{\mu-1}(x), \quad (1)$$

$$(2) \quad \frac{d}{dx} (x^{-\mu} J_\mu(x)) = -x^{-\mu} J_{\mu+1}(x). \quad (2)$$

$$\text{Recall } J_\mu(x) = \sum_{k=0}^{\infty} (-1)^k \frac{(x/2)^{2k+\mu}}{k! \Gamma(\mu + k + 1)}.$$

$$\begin{aligned} \text{Proof of (1): } \frac{d}{dx} (x^\mu J_\mu(x)) &= \frac{d}{dx} \left(\sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+2\mu}}{2^{2k+\mu} k! \Gamma(\mu + k + 1)} \right) \\ &= \sum_{k=0}^{\infty} (-1)^k \frac{2(k + \mu) x^{2k+2\mu-1}}{2^{2k+\mu} k! \Gamma(\mu + k + 1)} \\ &= \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+2\mu-1}}{2^{2k+(\mu-1)} k! \Gamma(\mu + k)} \\ &= x^\mu \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+(\mu-1)}}{2^{2k+(\mu-1)} k! \Gamma((\mu - 1) + k + 1)} = x^\mu J_{\mu-1}(x). \end{aligned}$$

In a similar manner, the result (2) can be obtained.

Opening up the differentiations on the LHS in equations (1) and (2) and multiplying them by $x^{-\mu}$ and x^{μ} , respectively, the following important results can be obtained:

$$J_{\mu-1}(x) + J_{\mu+1}(x) = \frac{2\mu}{x} J_{\mu}(x), \quad (3)$$

$$J_{\mu-1}(x) - J_{\mu+1}(x) = 2J'_{\mu}(x). \quad (4)$$

The above relations show the connection between some specific Bessel functions.

Specifically, for $\mu = n$,

Result (3) can be interpreted as the recursion relation between three consecutive Bessel functions of integer order whereas (4) shows that the derivative of an integer-order Bessel function can be expressed in terms of the Bessel functions of the preceding and following orders.

We know

$$J_{1/2}(x) = \sqrt{\frac{2}{\pi x}} \sin x,$$

$$J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}} \cos x.$$

Then, from (3), the following can be obtained:

For $\mu = 1/2$ and $\mu = -1/2$, we get, respectively,

$$J_{3/2}(x) = \frac{1}{x} J_{1/2}(x) - J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}} \left(\frac{\sin x}{x} - \cos x \right),$$

$$J_{-3/2}(x) = -\frac{1}{x} J_{-1/2}(x) - J_{1/2}(x) = \sqrt{\frac{2}{\pi x}} \left(-\frac{\cos x}{x} - \sin x \right).$$

For $\mu = 3/2$ and $\mu = -3/2$, we get, respectively,

$$J_{5/2}(x) = \frac{3}{x} J_{3/2}(x) - J_{1/2}(x) = \sqrt{\frac{2}{\pi x}} \left(\frac{3 \sin x}{x^2} - \frac{3 \cos x}{x} - \sin x \right),$$

$$J_{-5/2}(x) = -\frac{3}{x} J_{-3/2}(x) - J_{-1/2}(x) = \sqrt{\frac{2}{\pi x}} \left(\frac{3 \cos x}{x^2} + \frac{3 \sin x}{x} - \cos x \right).$$

- Such calculation can be continued indefinitely and hence, every Bessel function $J_{m+1/2}(x)$, with m as an integer, is elementary.
- Liouville proved that these are the only cases in which $J_\mu(x)$ is elementary. For all other values of μ , they are not elementary.

Definition of generating function:

Let $\{a_n\}_{n \geq 0}$ be a sequence of numbers. The generating function associated with this sequence is the series $G(x) = \sum_{n \geq 0} a_n x^n$.

In other words, the above series can be assumed to be generated by the function $G(x)$.

Consider the following and observe the generating functions:

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x},$$

$$\sum_{n=0}^{\infty} (-1)^n x^n = \frac{1}{1+x},$$

$$\sum_{n=0}^{\infty} (ax)^n = \frac{1}{1-ax},$$

$$\sum_{n=0}^{\infty} x^{2n} = \frac{1}{1-x^2}.$$

The generating function of the Bessel functions is

$$G(x, t) = \exp \left\{ \frac{1}{2} x \left(t - \frac{1}{t} \right) \right\}. \quad (5)$$

This function can be developed into a Laurent series. The coefficient of t^n in the expansion is the Bessel function of argument x and order n .

We can write

$$\exp \left\{ \frac{1}{2} x \left(t - \frac{1}{t} \right) \right\} = \sum_{n=-\infty}^{\infty} t^n J_n(x). \quad (6)$$

(6) can be obtained by considering the series expansions for $\exp(xt/2)$ and $\exp(-x/(2t))$, multiplying those series and then comparing the coefficients of t^n .

Further recall that

$$J_{-n}(x) = (-1)^n J_n(x). \quad (7)$$

Using (7), we can rewrite (6) as

$$\exp \left\{ \frac{1}{2} x \left(t - \frac{1}{t} \right) \right\} = J_0(x) + \sum_{n=1}^{\infty} [t^n + (-1)^n t^{-n}] J_n(x). \quad (8)$$

Let $t = \exp(i\theta)$ so that $\frac{1}{2}(t - 1/t) = \frac{\exp(i\theta) - \exp(-i\theta)}{2} = i \sin \theta$.

Then (8) gives

$$\exp(ix \sin \theta) = J_0(x) + \sum_{n=1}^{\infty} [\exp(in\theta) + (-1)^n \exp(-in\theta)] J_n(x). \quad (9)$$

We get

$$t^n + (-1)^n t^{-n} = [\exp(in\theta) + (-1)^n \exp(-in\theta)] = 2 \cos(2k\theta), \quad n = 2k, \quad (10)$$

$$t^n + (-1)^n t^{-n} = [\exp(in\theta) + (-1)^n \exp(-in\theta)] = 2i \sin((2k-1)\theta), \quad n = 2k-1. \quad (11)$$

Therefore, (9) can be written as

$$\exp(ix \sin \theta) = J_0(x) + 2 \sum_{k=1}^{\infty} J_{2k}(x) \cos(2k\theta) + 2i \sum_{k=1}^{\infty} J_{2k-1}(x) \sin((2k-1)\theta). \quad (12)$$

Equating real and imaginary parts,

$$\cos(x \sin \theta) = J_0(x) + 2 \sum_{k=1}^{\infty} J_{2k}(x) \cos(2k\theta), \quad (13)$$

$$\sin(x \sin \theta) = 2 \sum_{k=1}^{\infty} J_{2k-1}(x) \sin((2k-1)\theta). \quad (14)$$

Further, putting $\theta = \pi/2$ in (13) and (14), we get, respectively,

$$\cos x = J_0(x) + 2 \sum_{k=1}^{\infty} (-1)^k J_{2k}(x), \quad (15)$$

$$\sin x = 2 \sum_{k=1}^{\infty} (-1)^{k-1} J_{2k-1}(x). \quad (16)$$



Another important differential equation which gives rise to special functions (polynomials, to be specific) is **Legendre equation**:

$$(1 - x^2)y'' - 2xy' + n(n+1)y = 0, \quad \text{where } n \text{ is a positive integer.} \quad (17)$$

and its solutions are known as **Legendre polynomials**.

It is obvious that $x = \pm 1$ are regular singular points of Legendre equation.

We use regular power series $\sum_{k=0}^{\infty} a_k x^k$ (Frobenius series is not required) to find a series solution since **the origin is an ordinary point** in this case.

Because the region of convergence for such a series is $|x| < 1$, the infinite series will be convergent for $-1 < x < 1$.

Where do we get them?

When we use spherical coordinates (r, θ, ϕ) , the Laplace's equation $u_{xx} + u_{yy} + u_{zz} = 0$ gets converted to

$$u_{rr} + \frac{1}{r}u_r + \frac{1}{r^2}u_{\theta\theta} + \frac{\cot \theta}{r^2}u_\theta + \frac{1}{r^2 \sin^2 \theta}u_{\phi\phi} = 0. \quad (18)$$

After using the separation of variables, we come across three ordinary differential equations out of (18) one of which (in θ) is

$$(1 - x^2)y'' - 2xy' + \left[n(n+1) - \frac{m^2}{1 - x^2} \right] y = 0, \quad (19)$$

where $y = f(x)$ is a function of θ with $x = \cos \theta$.

This equation is known as **associated Legendre equation**, where m and n are constants with integral values, which arise while solving Laplace's equation by separation of variables. Its solutions are known as associated Legendre polynomials, denoted by P_n^m .

When $m = 0$, equation (19) reduces to Legendre equation (17).

Assume a series solution of the form

$$y = \sum_{k=0}^{\infty} a_k x^k. \quad (20)$$

Using (20), the derivatives $y' = \sum_{k=1}^{\infty} k a_k x^{k-1}$, $y'' = \sum_{k=2}^{\infty} k(k-1) a_k x^{k-2}$ in (17) and writing all series as $\sum_{k=0}^{\infty}$, we get

$$\sum_{k=0}^{\infty} a_k k(k-1) x^{k-2} + \sum_{k=0}^{\infty} a_{k-2} [n(n+1) - 2(k-2) - (k-2)(k-3)] x^{k-2} = 0. \quad (21)$$

Recurrence relation:

$$a_k = -\frac{[n(n+1) - 2(k-2) - (k-2)(k-3)]a_{k-2}}{k(k-1)}, \quad k \geq 2.$$

That is,

$$a_k = -\frac{[(n-k+2)(n+k-1)]a_{k-2}}{k(k-1)}, \quad k \geq 2. \quad (22)$$

The coefficients can be expressed in terms of either a_0 or a_1 as follows:

$$\begin{aligned} a_2 &= -\frac{n(n+1)}{2!}a_0, & a_3 &= -\frac{(n-1)(n+2)}{3!}a_1, \\ a_4 &= -\frac{(n-2)(n+3)}{4 \cdot 3}a_2 = +\frac{n(n-2)(n+1)(n+3)}{4!}a_0, \\ a_5 &= -\frac{(n-3)(n+4)}{5 \cdot 4}a_3 = +\frac{(n-1)(n-3)(n+2)(n+4)}{5!}a_1 \end{aligned}$$

A complete solution can be written in the form (in two power series)

$$\begin{aligned} y &= a_0 \left[1 - \frac{n(n+1)}{2!}x^2 + \frac{n(n-2)(n+1)(n+3)}{4!}x^4 - \dots \right] \\ &+ a_1 \left[x - \frac{(n-1)(n+2)}{3!}x^3 + \frac{(n-1)(n-3)(n+2)(n+4)}{5!}x^5 - \dots \right]. \end{aligned} \quad (23)$$

If n is even, the first series terminates after certain terms, and if n is odd, the second series terminates after certain terms. In either case, the series which reduces to a finite sum is known as a **Legendre polynomial** or a **spherical harmonic of order n** , denoted by $P_n(x)$.

To obtain the standard form of Legendre polynomial, we substitute the values of a_0 and a_1 in such a way that the coefficients of the highest power of x in each series is equal to

$$\frac{(2n)!}{2^n(n!)^2}$$

These values for a_0 and a_1 are

$$a_0 = (-1)^{n/2} \frac{n!}{2^n[(n/2)!]^2},$$
$$a_1 = (-1)^{(n-1)/2} \frac{(n+1)!}{2^n[(\frac{n-1}{2})!(\frac{n+1}{2})!]}.$$

The resulting general formula is

$$P_n(x) = \sum_{k=0}^N \frac{(-1)^k (2n-2k)!}{2^n k! (n-k)! (n-2k)!} x^{n-2k}, \quad (24)$$

where

$$N = \frac{n}{2}, \quad n \text{ even}; \quad N = \frac{n-1}{2}, \quad n \text{ odd}.$$

This formula gives the first few Legendre polynomials as:

$$P_0(x) = 1, P_1(x) = x, P_2(x) = \frac{1}{2}(3x^2 - 1), P_3(x) = \frac{1}{2}(5x^3 - 3x),$$

$$P_4(x) = \frac{1}{8}(35x^4 - 30x^2 + 3), P_5(x) = \frac{1}{8}(63x^5 - 70x^3 + 15x),$$

$$P_6(x) = \frac{1}{16}(231x^6 - 315x^4 + 105x^2 - 5), P_7(x) = \frac{1}{16}(429x^7 - 693x^5 + 315x^3 - 35x),$$

$$P_8(x) = \frac{1}{128}(6435x^8 - 12012x^6 + 6930x^4 - 1260x^2 + 35).$$

A complete solution of (17) can be written as

$$y(x) = AP_n(x) + BQ_n(x), \quad (25)$$

where $Q_n(x)$, the Legendre polynomial of second kind can be obtained from $P_n(x)$ as

$$Q_n(x) = P_n(x) \int \frac{dx}{(x^2 - 1)[P_n(x)]^2} \quad (26)$$

which is not bounded near $x = \pm 1$.

Legendre equation of order zero ($n = 0$) can be written as

$$(1 - x^2)y'' - 2xy' = 0, \quad (27)$$

which has the compact form

$$\frac{d}{dx} \left[(1 - x^2) \frac{dy}{dx} \right] = 0. \quad (28)$$

Integrating (28):

$$(1 - x^2) \frac{dy}{dx} = A. \quad (29)$$

Separating the variables and integrating (29):

$$\int dy = \int \frac{A}{1 - x^2} dx.$$

This gives the solution as

$$y = A \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right) + B. \quad (30)$$

We know that, being of second order, every Legendre equation has two linearly independent solutions with two arbitrary constants which are A and B here.

Therefore, here we denote the linearly independent solutions of (27) as

$$P_0(x) = 1, \quad Q_0(x) = \frac{1}{2} \ln \left(\frac{1+x}{1-x} \right). \quad (31)$$

It may be noted that $Q_0(x)$ diverges at $x = 1$.

As it has already been observed, for the general case of $n \neq 0$, power series solution is employed to solve Legendre equation.

Note:

In most of the cases, since solutions bounded for $x \rightarrow \pm 1$ are required, it is seen that the solution of a specific Legendre equation of order n tends to **contain only $P_n(x)$** , discarding $Q_n(x)$.

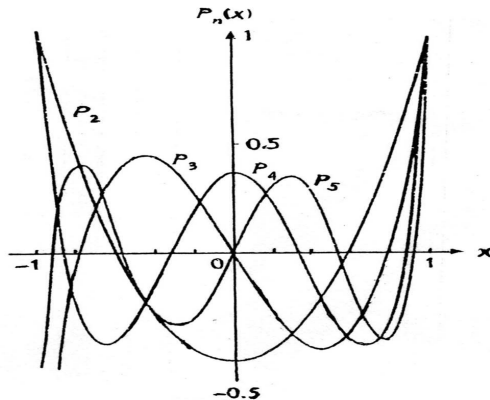


Figure 1 : Graphs of some Legendre polynomials